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Bridging the gap between extension of time prediction and contract administration: an explainable ensemble model for delay analysis under FIDIC contracts

Mohammad Taghipour [a], Mostafa Khanzadi [a], Abdollah Amirkhani [b,c,*]

[a] School of Civil Engineering, Iran University of Science and Technology, Tehran 16846-13114, Iran
[b] School of Automotive Engineering, Iran University of Science and Technology, Tehran 16846-13114, Iran
[c] School of Computer Engineering, Iran University of Science and Technology, Tehran 16846-13114, Iran

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ABSTRACT

Construction schedule overruns remain a systemic crisis. This study introduces a unified predictive framework to forecast the Duration of Delay Fragment (DDF), a novel granular metric decoupling technical time requirements from contractual negotiation. By integrating a Stacking Ensemble model, combining Random Forest, XGBoost, LightGBM, and CatBoost, with a Multi Criteria Decision Analysis interpretation layer, the research addresses both predictive accuracy and legal explainability. Trained on 1,594 delay events from projects governed by FIDIC 2017 Suite of Contracts within the MENA region, the model achieved a Test R^2 of 0.817 and normalized RMSE of 0.445, demonstrating enhanced predictive performance relative to the individual baseline algorithms. Shapley Additive Explanations (SHAP) analysis revealed that Type of Excusable Delay (TXD) and Network Interdependency Index (NII) are critical multipliers of delay magnitude. The proposed framework provides contract administrators with a scientifically grounded, transparent tool for objectively validating Extension of Time (EOT) claims.

1. Introduction

The timely delivery of construction projects constitutes a fundamental performance indicator for stakeholders, as it directly governs the financial viability of investments [1]. Despite the widespread adoption of standardized frameworks, schedule overruns remain a systemic and pervasive crisis with recent empirical studies highlighting the high percentage of schedule overruns in construction projects, particularly in developing regions and complex contractual environments [2]. These deviations from the planned schedule result in severe financial consequences, including substantial cost escalations, the imposition of liquidated damages, and prolonged legal disputes [3]. While traditional deterministic methods such as the Critical Path Method (CPM) are contractually mandated for scheduling, research demonstrates that these techniques often fail to capture the complex, nonlinear interactions between risk factors, rendering them insufficient for accurate duration prediction in dynamic project environments [4]. Furthermore, existing administrative protocols for delay analysis, often reliant on manual heuristics, struggle to process the high volume of unstructured site data, required to substantiate claims within the strict notice periods mandated by standard contracts such as FIDIC 2017 Suite of Contracts [5–8].

To mitigate these risks, the industry has historically relied on statistical control mechanisms such as Earned Value Management (EVM) or Earned Schedule Method (ESM) to forecast project outcomes. However, these methods are predicated on the linear extrapolation of historical performance indices, a theoretical assumption that has been proven inadequate for capturing the nonmonotonic impact of disruptive events [9]. Recent studies have further highlighted that the combined impact of multiple risk factors is often nonadditive, meaning that simple summation models fail to account for the "superposition effect" where interacting risks create nonlinear delays [10]. Consequently, there is a consensus in recent literature that static methods are insufficient for modern construction challenges, necessitating a shift toward more dynamic, automated solutions [11,12].

In response to these methodological limitations, the domain has pivoted toward Machine Learning (ML) as superior alternatives for predictive modeling [13]. The literature demonstrates a clear trajectory

* Corresponding author at: School of Automotive Engineering, Iran University of Science and Technology, Tehran 16846-13114, Iran.
E-mail addresses: m_taghipour@civileng.iust.ac.ir (M. Taghipour), khanzadi@iust.ac.ir (M. Khanzadi), amirkhani@iust.ac.ir (A. Amirkhani).

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Table 1
Comparative analysis of delay prediction models in construction management literature.

Output Type	Study	Algorithm/Method	Inputs (Features)	Output (Target)	Performance
Binary Classification (Delayed / Not Delayed)	Fan [15]	Multilayer Perceptron (MLP)	Construction quality data, schedule performance metrics.	Schedule Status (Ahead/Behind)	Accuracy: 98.4%
	Santos et al. [21]	Gradient Boosting Classifier	Task durations, Stochastic distributions from Monte Carlo simulation.	Compliance (On Time / Late)	Accuracy: 85.7%
	Egwim et al. [30]	Ensemble Model	9 optimized risk factors (e.g., equipment breakdown, inflation, communication).	Project Delay (0 or 1)	Accuracy: 80.65%
	Asadi et al. [31]	J48 Decision Tree	11 quantitative attributes (e.g., weather, manpower) + 49 qualitative survey factors.	Delay Status (Yes/No)	Accuracy: 79.41%
	Egwim & Alaka [20]	Perceptron	9 factors: (e.g., Design variations, price fluctuation).	Project Delay (0 or 1)	Accuracy: 85%
	Awada et al. [32]	Random Forest (RF)	Onsite data stream (CPRs).	Delay Status (Yes/No)	Accuracy: 91.1%
Multi-class Classification (Delay Severity Levels)	Shirazi & Toosi [33]	Deep MLP + PCA	5 Principal Components (derived from 65 risk factors).	Delay Level (<50%, 50–100%, >100%)	Accuracy: 94.36%
	Sanni-Anibire et al. [34]	Artificial Neural Network (ANN)	Decision slowness, Subcontractor delay, Late instructions, Shop drawing approval.	Risk Level (Moderate, High, Very High)	Accuracy: 93.75%
	Gondia et al. [35]	Naïve Bayes	9 Risk Sources (e.g., Consultant, Contractor).	Delay Extent (<30%, 30–60%, >60%)	Accuracy: 51.2%
	Yaseen et al. [36]	Hybrid RF − GA	8 Delay Sources (e.g., Consultant, Contractor).	Delay Level (<50%, 50–100%, >100%)	Accuracy: 91.67%
Regression Prediction (Delay Amount/ Deviation)	Shihadeh et al. [37]	MLP- CROA (Optimized Deep Learning)	Planned Cost, Actual Cost, Planned Duration, Actual Duration, Cost Overrun.	Time Overrun (Days)	R^2: 0.991
	Fitzsimmons et al. [38]	Hybrid (SVM + Monte Carlo)	Task Descriptions (NLP), Concurrent Activity Level, Total Float, Weather, Economic Conditions.	Task Duration Deviation	Std Error: 16.7 weeks
	Yang & Chen [39]	Enhanced ESM	Planned Duration, Actual Time, Baseline Slack, Start Date, Percent Complete.	Estimated Time at Completion	High Precision
	Senić et al. [40]	Sugeno Fuzzy Logic	10 Characteristics (e.g., Contract Amount, Time).	EOT %	Mean Deviation: 10.2%
	Son & Lee [41]	Text Mining + Naïve Bayes	Unstructured Contract Text.	Time Overrun (Months)	Accuracy: 80.7
Time Series & Forecasting (Progress Over Time)	Rudeli et al. [42]	Markov Chains	Gradual Progress Percentage (Current vs Previous state).	Total Schedule Deviation	Mean Deviation: 4.29%

from simple statistical regression, which often explains less than 70% of variance, to high dimensional ML models capable of capturing complex nonlinear interactions between project variables [14]. This shift is supported by the increasing availability of digital project data, which allows for the training of robust classifiers capable of identifying latent risk patterns that human analyst might overlook [15,16]. Models such as Long Short-Term Memory (LSTM) networks have proven exceptionally effective at modeling sequential construction data, achieving prediction accuracies exceeding 94% in complex environments [17]. Surveys of state-of-the-art forecasting methods confirm that Hybrid and Ensemble approaches currently represent the peak of predictive capability [18].

However, despite these computational advancements, a critical "implementation gap" persists between academic algorithms and practical contract administration [19], driven by three fundamental limitations in the current body of knowledge. First, the majority of existing studies focus on predicting the total project duration or binary delay status, which, while useful for macro forecasting, lacks the granularity required for forensic delay analysis of specific claim events [20]. Second, the "Black Box" nature of advanced ML models hinders their adoption in dispute resolution, where explainability is a legal necessity; a prediction without an explicable causal trace is often inadmissible in arbitration [21,22]. While recent efforts have employed visualizers or Game Engines to improve communication, the underlying interpretability of the quantitative models remains a challenge [23]. Third, most predictive models rely heavily on quantitative project metrics while underutilizing the specific contractual and qualitative risk signatures such as "unforeseeable physical conditions" that legally define delay entitlement under FIDIC [24].

Recent advancements in forensic delay theory have increasingly emphasized the need to distinguish between the technical duration of an event and the legally compensable outcome that emerges after the application of contractual rules. This distinction is embedded in the Society of Construction Law (SCL) [25] and AACE International Recommended Practice [26], both of which separate the "cause-level" technical effect of an event from the subsequent entitlement determination. Bektas et al. [27] formalized this bifurcation by characterizing Time Impact Analysis (TIA) as the domain of technical causation, whereas dispute resolution processes determine contractual liability. Likewise, Zhang & Li [28] and Saini & Thomas [29] demonstrated that prediction models improve substantially when delay is decomposed into granular, fragment-level components rather than global project durations.

Building on these theoretical foundations, this study introduces the Duration of Delay Fragment (DDF) as a formally defined construct representing the intrinsic duration of a TIA, independent of float consumption, concurrency handling, apportionment rules, or Extension of Time (EOT) negotiation. Unlike traditional retrospective analysis, where fragment durations are often subjectively adjusted to align with administrative targets, DDF isolates the pure technical time required by the event itself and captures the causal reality of delay events rather than their contractually modified outcome.

The proposed framework utilizes a combination of contractual and engineered features to estimate the DDF as output. The input variables include contractual descriptors such as Project Delivery Method (PDM), Project Payment Method (PPM), Initial Contract Cost (ICC), Initial Project Time (IPT), Contract Signing Date (CSD), and Type of Excusable Delay (TXD). In addition, engineered features include Affected Duration Magnitude (ADM), Cost Density Ratio (CDR), Critical Time Ratio (CTR), Minimum Remaining Float (MRF), Network Interdependency Index (NII), and Prior Delay Density (PDD).

This study aims to bridge the introduced gap by developing a predictive framework. The principal contributions of this study are summarized as follows:

- The current study proposes a novel unit (DDF) which isolates the duration of individual delay factors from the final negotiated delay outcomes.

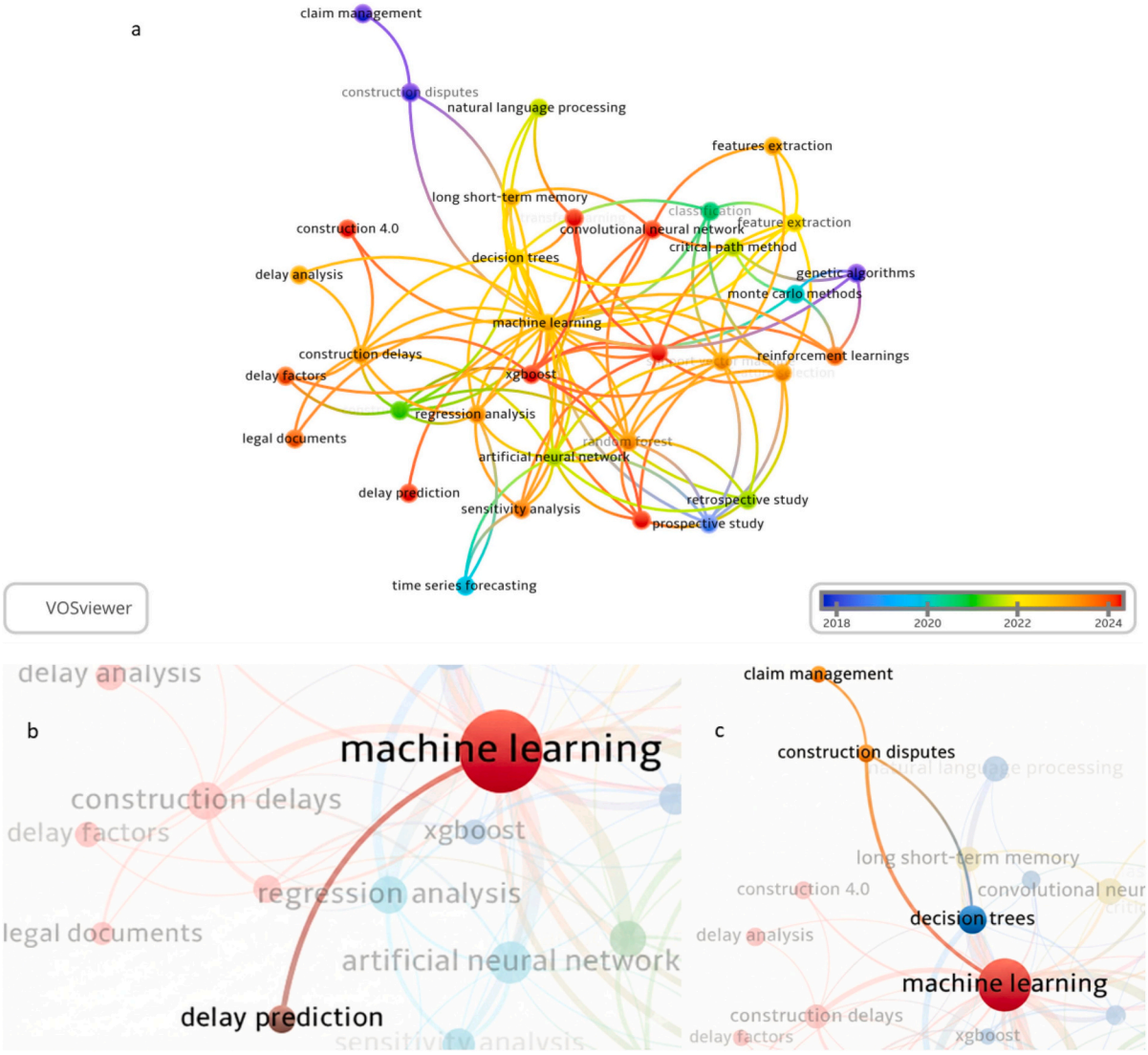


Fig. 1. Scientometric analysis of the research domain evolution. (a) Temporal overlay visualization of keyword cooccurrences (b) Magnified view of the Delay Prediction node (c) Magnified view of the Construction Claim node.

- The study establishes a robust link between forensic delay variables and the FIDIC 2017 Suite of Contracts, providing a standardized predictive tool for contractors, consultants, and contract administrators to objectively evaluate EOT claims
- Development of a predictive architecture combining forensic TIA with advanced Ensemble learning models and engineering of features that quantify delay magnitude, schedule constraint, and network complexity.
- By integrating a Multi-Criteria Decision Analysis (MCDA) layer using Shapley Additive Explanations (SHAP) with the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) method, the study transforms "black-box" ML into a transparent tool.

2. Literature review

Delay prediction in construction schedule management is characterized by two distinct methodological approaches: categorical classification and quantitative regression. Table 1, shows studies that investigate delay prediction in construction area. Classification models have achieved high accuracies in determining binary schedule status. While effective for early warning, these categorical outputs lack the precise time quantification necessary for establishing contractual entitlement in forensic delay analysis. Consequently, researchers have developed regression to forecast continuous duration variables.

A notable outlier in this domain is the study by Shihadeh et al. [37], which reported an exceptional Coefficient of Determination (R^2) of 0.991 using an optimized MLP. Nevertheless, the reported predictive performance should be interpreted cautiously in the context of forensic delay forecasting. The model incorporates variables such as "Actual Cost" and "Actual Duration" as predictive inputs, despite these parameters becoming fully observable only after project execution and exhibiting strong dependency relationships with the target variable (time overrun). The inclusion of such post-factum variables introduces a significant risk of data leakage, whereby information indirectly associated with the prediction target becomes embedded within the input space. Although these variables may be appropriate for retrospective project assessment, their applicability in prospective EOT evaluation is comparatively limited.

To empirically map the evolutionary trajectory of the domain, a keyword cooccurrence analysis was conducted on the metadata of high impact publications using VOSviewer. The temporal overlay visualization, illustrated in Fig. 1(a), reveals a distinct chronological stratification of research themes. Early research efforts, were predominantly focused on "genetic algorithms", reflecting an era where delay analysis

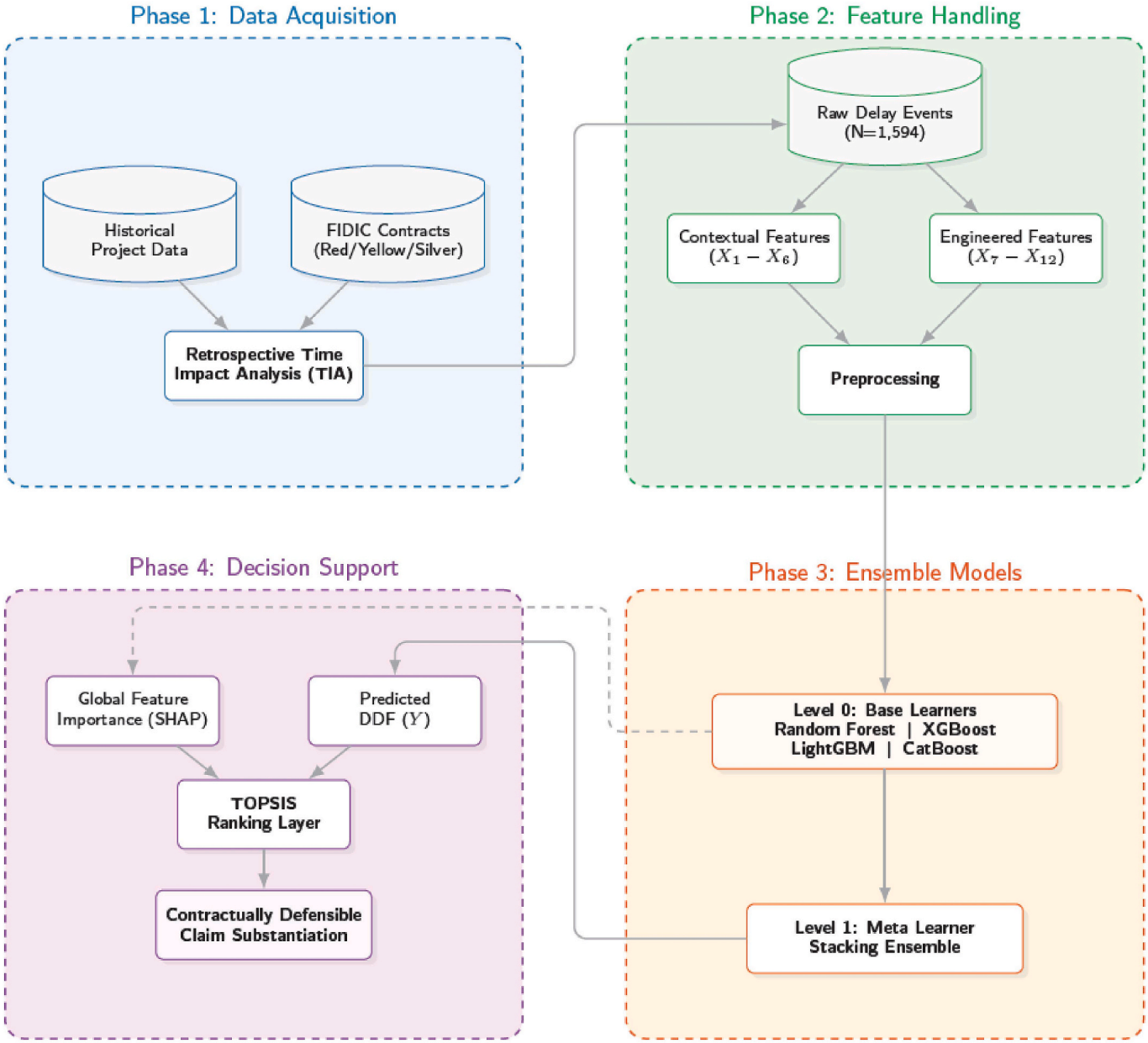


Fig. 2. The proposed methodological framework.

was largely optimization based. A transitional phase is observed between 2020–2021, characterized by stochastic methodologies such as "Monte Carlo Methods" and "Fuzzy Logic", which sought to address uncertainty but lacked high dimensional predictive capacity.

In sharp contrast, the most recent cluster is dominated by "Delay Prediction" and "Construction 4.0″, confirming a rapid paradigm shift toward automated intelligence. Most notably, "XGBoost" appears as one of the newest nodes validating the selection of Ensemble models in the field of construction engineering as the current state-of-the-art [43,44]. As detailed in Fig. 1(b), the term "Delay Prediction" is densely connected to the "Machine Learning" node indicating a pure technical system. However, Fig. 1(c) reveals that "Claim Management" exists as a peripheral, isolated node with sparse connectivity. This visual disconnection indicates that while predictive technologies have advanced, they have not yet been integrated into the forensic frameworks used for claim substantiation.

3. Research methodology

To achieve the study objectives, a unified predictive framework was developed. The overall architectural flow of the research methodology is illustrated in Fig. 2. 38 projects from the MENA region were included in this study based on three inclusion criteria: (1) execution under a FIDIC 2017 Suite of Contracts, (2) project value exceeding $3.5 million, and (3) possession of a complete, logically linked baseline schedule.

3.1. Contractual framework and data strategy

The predictive modeling exercise focuses on construction projects administered under the FIDIC 2017 Suite of Contracts [6–8]. Adherence to this standardized framework is essential for establishing consistency in classifying contractual risks and determining the basis for time entitlements. Under this framework, the adjustment of the project completion date is governed by the EOT provision (Sub Clause 8.5). To isolate the technical cause from the contractual consequence, this study

Table 2
Taxonomy of delay events and resultant entitlements under FIDIC 2017 suite.

Delay Type	Definition	Resultant Entitlement	Primary Risk Bearer
Compensable	Delay caused by a breach, instruction, or action/omission of the Employer or Engineer.	EOT (Extension of Time) + Cost (and often profit)	Employer
Excusable None compensable	Delay caused by neutral, external events that are neither party's fault.	EOT only (Time, but no compensation for prolongation costs)	Neutral (Risk allocated to Employer)
None Excusable	Delay caused by the Contractor's own inefficiencies, errors, or default.	No EOT, No Cost (Contractor liable for liquidated damages)	Contractor

Table 3
Classification of delay factors and associated contractual sub-clauses.

Number	Delay Factor (FIDIC Terminology)	Fidic Sub-Clause	Entitlement
1	A Variation instructed or an adjustment to the Accepted Contract Amount.	8.5(a) & 13.3	Time & Cost
2	The Employer failing to give the Contractor right of access to or possession of the Site.	2.1	Time & Cost
3	A delay in the issue of any drawing or instruction required for the execution of the Works.	1.9	Time & Cost
4	Unforeseeable Physical Conditions (excluding climatic conditions).	4.12	Time & Cost
5	Compliance with the Engineer's instructions regarding fossils, coins, or items of historical or archaeological interest.	4.24	Time & Cost
6	Suspension of the Works by the Employer not due to the Contractor's default.	8.8	Time & Cost
7	Errors in the Employer's Requirements which an experienced contractor could not reasonably have discovered.	1.9	Time & Cost
8	Any cause of delay giving an entitlement to EOT under a Sub-Clause other than Sub-Clause 8.5 (e.g., Sub-Clause 17.3 for Employer's risks such as War and Riot).	8.5(b)	Time & Cost
9	Exceptionally adverse climatic conditions.	8.5(c)	Time Only
10	Unforeseeable shortages in the availability of personnel or Goods caused by epidemic, government actions, or unforeseeable scarcity.	8.5(d)	Time Only
11	Unforeseeable delay, impediment, or prevention caused by or attributable to any public authority.	8.5(e)	Time & Cost
12	An Exceptional Event (Force Majeure).	18.1—18.4	Time Only
13	Any delay, impediment or prevention caused by or attributable to the Employer or the Employer's Personnel.	8.5(g)	Time & Cost
14	Suspension of work due to the Employer's failure to pay the Contractor.	16.1	Time & Cost

introduces the DDF. The DDF is the primary object of prediction (Y) because it represents the core causal time quantity inherent to the change itself, thereby decoupling the model's output from the complex mechanics of the schedule and subjective contractual bias.

In the context of the FIDIC 2017 Suite of Contracts, the determination of a valid claim for time relies on the contractual assignment of risk for the delaying event. Delays are fundamentally categorized by the resulting contractual entitlement for the contractor represented in Table 2.

All excusable delay events, regardless of their financial entitlement (cost or no cost), are relevant inputs for predicting the DDF. The model's objective is to predict the technical time block required by the event; consequently, both compensable and excusable none compensable events are included in the dataset, as both lead to a nonzero DDF. The excusable events that grant an EOT under FIDIC 2017 contracts are classified in Table 3.

A concurrent delay arises when the project's critical path is simultaneously affected by two or more independent events, where at least one is a contractor risk and at least one is an employer risk. While the legal assessment typically determines that the contractor is entitled to the EOT but not the associated prolongation costs, the key methodological distinction for this study is: The model includes all excusable events because the magnitude of their technical time requirement (DDF) is the critical predictive factor, independent of the financial outcome.

The process of acquiring the factual and causal data required for model training demands a highly robust and justifiable delay analysis method. Delay analysis methods are broadly categorized based on their temporal application: prospective (forecasting impact before execution) or retrospective (forensically analyzing execution records). The selection of the technique is governed by the need to ensure factual accuracy and establish the pure causal link required for the ML model. The Table 4 compares the five principal methods based on industry standards, including references to the SCL Protocols [25] and the AACE International Recommended Practices [26].

Based on the necessity to establish a factual, causal, and dynamic relations between the delay event and its duration, the retrospective TIA protocol is adopted for data acquisition. This method provides the required granularity for the ML features. The analysis is executed to ensure the output of model is a measure of the pure technical impact of the employer's delay fragment, adhering to the following protocol:

- Establish the “But-For” Schedule: The contemporaneous schedule update immediately preceding the delay event is used to define the exact project status and critical path.
- Model and measure the DDF: The excusable events are modeled as a new delay fragment and inserted into the “but-for” schedule. The time allocated to this inserted fragment is the direct measurement for the target variable Y.
- Preventing apportionment bias: The analysis deliberately records the DDF as the full, gross technical time of the employer’s delay fragment, ignoring any concurrent time apportionment rules. This ensures the resulting dataset is a measure of the event's technical severity, free from legal negotiation or contractual dilution.

3.2. Dataset construction

The construction of the predictive dataset relies on extracting factual metrics from the historical project records via the retrospective TIA protocol. The data is categorized into the target variable, contextual variables, and granular source variables used for subsequent feature engineering.

3.2.1. Raw data variables

Contextual variables are extracted from the contractual documents and the master project plan, defining the risk environment and scale of the project instance.

X_1 = PDM: The contractual mechanism governing project execution, codified as a categorical variable including:

- Design-Bid-build (DBB)
- Construction Manager (CM)
- Design-Build (DB)
- Engineering, Procurement, and Construction (EPC) / Turnkey
- Public Private Partnership (PPP)

Table 4
Comparative analysis of forensic delay analysis methodologies for data acquisition.

Category	Method	Reference	Advantages	Disadvantages
Prospective	Impacted As-Planned (IAP)	AACE RP 29R-03 (Method 3.1)	Quick; isolates single events theoretically.	Lacks factual basis; ignores actual progress; rarely accepted for final claims.
Prospective /Retrospective	Time Impact Analysis (TIA)	SCL Protocol (Section 4.4)	Factual, causal, and dynamic. Uses contemporaneous schedules to accurately measure the impact of an event on the current critical path.	requires a logically linked schedule record.
Retrospective	As-Planned vs. As-Built (APAB)	SCL Protocol (Section 4.5)	Simple, visually effective; establishes total slippage (factual duration difference).	Fails to establish causation or criticality; cannot isolate individual DDFs; primarily a screening tool.
Retrospective	Collapsed As-Built (CAB)	AACE RP 29R-03 (Method 3.6)	Factual and isolates liability; determines EOT net of Contractor delays; highly persuasive in dispute resolution.	Extremely data intensive; computationally complex.
Retrospective	Window Analysis (WA)	SCL Protocol (Section 4.6)	Robust. Sequential TIA, best for complex, long duration projects and analyzing concurrent delays.	Most resource intensive; requires a series of accurate schedule updates across the timeline.

Table 5
Allowable excusable delay factors by PDM.

Type of Excusable Delay	DBB (Fidic Red)	CM (Fidic Red)	DB (Fidic Yellow)	EPC/ Turnkey (Fidic Silver)	PPP (Fidic Silver/ Gold)
Variations (Changes in Scope)	✔	✔	✔	✔	✔
Delayed Site Access	✔	✔	✔	✔	✔
Delayed Instructions/ Drawings	✔	✔	✔	✔ (Limited)	✔ (Limited)
Unforeseeable Physical Conditions	✔	✔	✔	✘	✘
Archaeological Finds (Fossils)	✔	✔	✔	✔	✔
Suspension of Works (Employer Order)	✔	✔	✔	✔	✔
Errors in Employer's Requirements	✔	✔	✔	✘	✘
Any Other Type of Delays (Employer's Side)	✔	✔	✔	✔	✔
Exceptionally Adverse Climatic Conditions	✔	✔	✔	✘	✘
Unforeseeable Shortages (Epidemics/Govt Action)	✔	✔	✔	✘	✘
Delays by Authorities (Unforeseeable)	✔	✔	✔	✔	✔
Exceptional Event (Force Majeure)	✔	✔	✔	✔	✔
Delays/ Interference by Employer Personnel	✔	✔	✔	✔	✔
Delay on Financial Payments (Suspension)	✔	✔	✔	✔	✔

Table 5 represents what type of excusable delay factor allowed within the scope of each PDM.

X_2 = PPM: The contractual payment structure which serves as a primary proxy for the allocation of commercial risk including:

- Lump-Sum (fix price)
- Unit Price
- Cost Plus

Table 6
Categorization of MRF and associated time constraint levels.

Category Label	Range of MRF (days)	Interpretation
A	0	Critical: The highest risk level. Any delay impacts project completion.
B	1–5	Near Critical: Extremely high constraint; a minimal buffer remaining. High urgency for DDF resolution.
C	6–15	Midrange Constraint: Moderate time pressure; typical range for close monitoring.
D	15 <	Low Constraint/Noncritical: Sufficient buffer to absorb minor delays, indicating low time pressure.

- Guarantied Max Price

X_3 = ICC: The total planned project budget ($).

X_4 = IPT: The total planned duration stated in the contract program (days).

X_5 = CSD: The date the contract was formally executed.

X_6 = TXD: The specific contractual cause of the delay mentioned in Table 3.

Source variables (X_{raw}) are the fundamental, granular metrics extracted via TIA for every activity affected by the excusable delay event. An activity is considered affected if it is either the recipient of the excusable delay event or if its planned scope or execution logic must be modified to accommodate the execution of the delay fragment. These raw variables are later aggregated and transformed to create the model's complex predictive features.

$D_{affected}$ = The planned working duration of the individual affected activity as recorded in the contemporaneous "but-for" schedule.

$C_{affected}$ = The cost assigned to the individual affected activity. This value is standardized to a single currency across the entire dataset.

$F_{affected}$ = The float of the individual affected activity in the "but-for" schedule. This measures the available time buffer before the activity affects the project completion date.

$I_{affected}$ = The count of immediate logical predecessor and successor links assigned to the individual affected activity, capturing its connectivity within the schedule network.

$N_{affected}$ = The number of total affected activities by an individual excusable delay event.

3.2.2. *Feature engineering*

The predictive strength of the model is rooted in the engineering of features that quantify the three principal drivers of delay: Magnitude (X_7-X_8), Constraint (X_9-X_{10}), and Complexity (X_{11}-X_{12}). These features are derived by aggregating and transforming the standardized source variables X_{raw} extracted from the set of affected activities called $S_{affected}$.

X_7 = ADM: Calculated as the simple sum of the planned durations of all activities affected by the delay via Eq. (1). This feature represents the total planned timescale of the impacted scope.

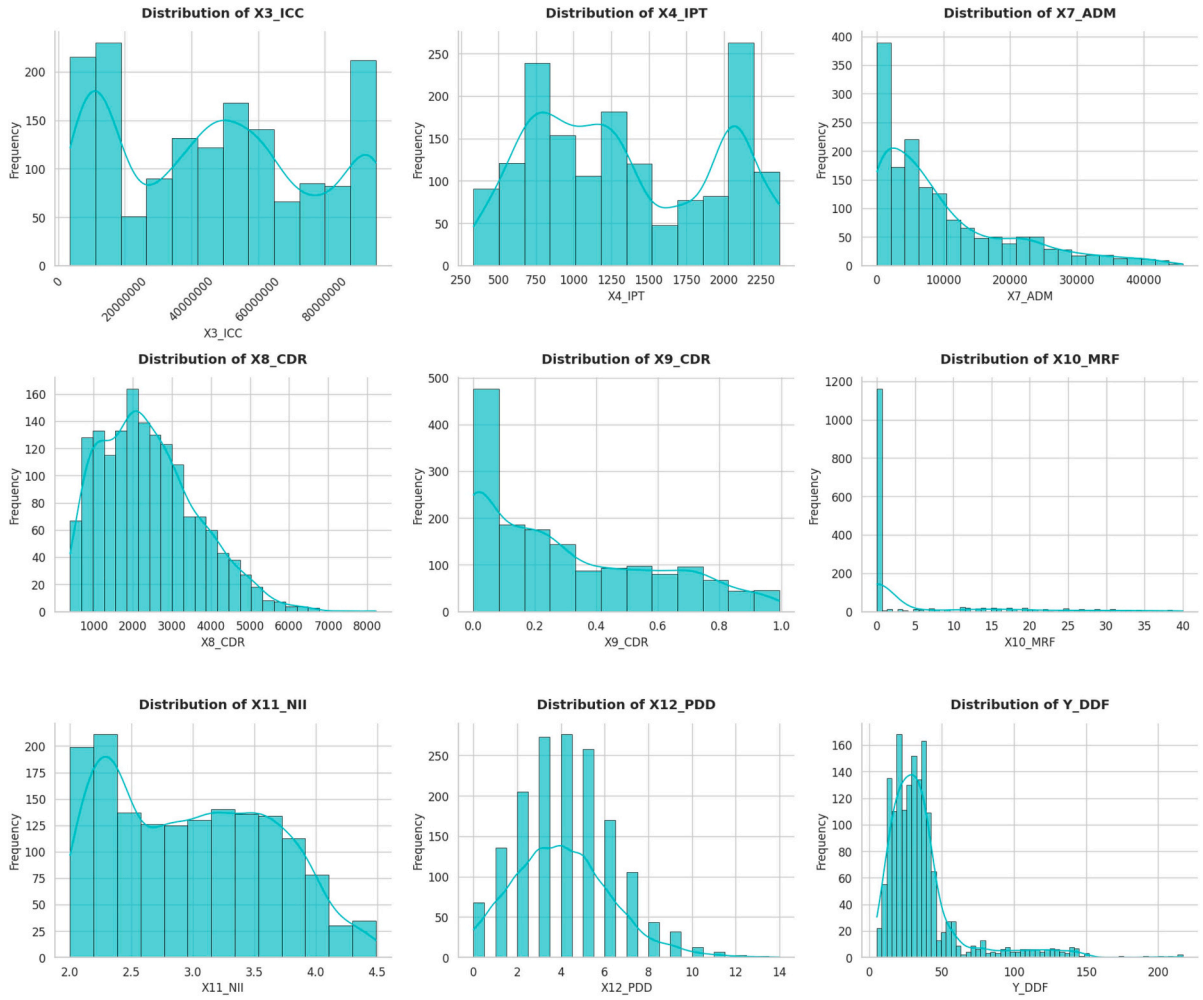


Fig. 3. Distribution of numerical feature values across the dataset.

$$X_7 = \sum_{i \in S_{affected}} D_i \tag{1}$$

X_8 = CDR: Quantifies the resource intensity of the affected scope via Eq. (2). It is computed as the ratio of total cost to total duration of affected activities. This ratio is critical because it transforms raw cost data into a measure of complexity and resource concentration (Cost per Day), accurately distinguishing between low density (easy to change) and high density (difficult to redesign) work, which directly influences the required DDF

$$X_8 = \frac{\sum_{i \in S_{affected}} C_i}{\sum_{i \in S_{affected}} D_i} \tag{2}$$

X_9 = CTR: Measures the density and spread of criticality within the affected scope via Eq. (3). It is calculated as the ratio of the total planned duration of the critically affected activities to the total duration of all affected activities (X_7). This feature solves the zero-float saturation problem, allowing the model to distinguish between a single critical activity and a pervasive, project wide critical bottleneck.

$$X_9 = \frac{\sum_{i \in S_{affected}, F_i = 0} D_i}{X_7} \tag{3}$$

X_{10} = MRF: Derived from the minimum float observed among all affected activities. This value is converted into a categorical to capture the gradient of time pressure. This is essential for modeling the urgency of delay mitigation, differentiating between a high risk, near critical delay and a negligible risk delay as it represented in Table 6.

X_{11} = NII: Calculated as the average count of logical predecessors and successors for all affected activities via Eq. (4). A higher index indicates greater schedule integration complexity, which necessitates increased managerial review and planning time, thus driving up the DDF.

$$X_{11} = \frac{\sum_{i \in S_{affected}} I_i}{|N_{affected}|} \tag{4}$$

X_{12} = PDD: A temporal feature calculated as the total count of excusable delay events that occurred within a defined lookback period which is 90 days, immediately preceding the current delay event. This feature quantifies the project instability and distress, modeling the fact that changes in a reactive, unstable environment require greater DDF for administrative processing and schedule stabilization.

It is important to emphasize that all features employed within the framework were designed to preserve prospective applicability and prevent data leakage during prediction. All engineered features derived from the source variables. In addition, contextual and source variables were constructed exclusively using information available at the time of claim evaluation, including contractual records, baseline schedules, and schedule updates.

3.2.3. *Data preprocessing*

The raw data must be rigorously preprocessed to ensure input features are consistent, stable, and suitable for nonlinear regression modeling. This involves addressing data integrity first, followed by transformation and final normalization.

Table 7
Distribution of projects and delay events by contractual attributes.

Feature Category	Variable Class	Project Count (Out of 38)	Project %	Delay Count (Out of 1594)	Delay %
Project Delivery Method (X1)	PPP (Public-Private Partnership)	7	18.40%	534	33.50%
	EPC (Engineering, Procurement, Construction)	7	18.40%	439	27.50%
	DBB (Design-Bid-Build)	12	31.60%	384	24.10%
	DB (Design-Build)	7	18.40%	136	8.50%
	CM (Construction Management)	5	13.20%	101	6.40%
Project Payment Method (X2)	GMP (Guaranteed Maximum Price)	7	18.40%	534	33.50%
	Lump Sum	14	36.80%	575	36.10%
	Unit Price	12	31.60%	384	24.10%
	Cost Plus	5	13.20%	101	6.30%
Type of Excusable Delay (X6)	Variations (Changes in Scope)			454	28.48%
	Delayed Instructions / Drawings			445	27.92%
	Any Other Type of Delays (Employer's Side)			154	9.66%
	Delays by Authorities			134	8.41%
	Suspension of Works			90	5.65%
	Delays/Interference by Employer Personnel			79	4.96%
	Errors in Employer's Requirements			71	4.45%
	Delay on Financial Payments			35	2.20%
	Delayed Site Access			32	2.01%
	Exceptionally Adverse Climatic Conditions			31	1.94%
	Unforeseeable Shortages			28	1.76%
	Unforeseeable Physical Conditions			26	1.63%
	Force Majeure (Exceptional Event)			15	0.94%

Data integrity checks are performed initially to prevent skewing of subsequent statistical analysis:

- Missing Values: Any missing raw values are addressed using statistical imputation techniques, provided the rate of missingness does not compromise the validity of the data point.
- Outlier Detection: Extreme outliers in raw numerical features are identified using the Interquartile Range (IQR) method. Outliers are appropriately removed to prevent them from unduly influencing the mean and standard deviation, which would degrade the subsequent normalization steps.

Once data integrity is verified, features are prepared for the ML environment:

- Categorical Encoding: All nominal variables X_1 (PDM), X_2 (PPM), X_6 (TXD), X_{10} (RMF) are transformed using One-Hot Encoding. This prevents the model from assuming a false ordinal relationship between distinct contract or delay types.
- Date Conversion: The raw date X_5 (CSD) is converted into numerical features to facilitate temporal analysis.
- Scale Normalization: Continuous numerical features, including the final engineered features and the target, are scaled using Z-score normalization to ensure uniform input distributions and prevent variables with naturally large raw values from dominating the model optimization process.

3.3. ML algorithms

This study adopts Ensemble learning as the foundation for the predictive architecture. Ensemble learning is a ML paradigm where multiple individual models, often referred to as "weak learners" (typically decision trees), are strategically combined to obtain a superior predictive performance than any single model could achieve alone. The

Table 8
Representative sample of the final training dataset.

Event ID	Project ID	X1 (PDM)	X2 (PPM)	X3 (ICC)	X4 (IPT)	X5 (CSD)	X6 (TXD)	X7 (ADM)	X8 (CDR)	X9 (CTR)	X10 (MRF)	X11 (NII)	X12 (PDD)	Y (DDF)
EV-0008	P01	DBB	Unit Price	14,242,834	596	4/8/2019	Delayed Instructions	360	1117.19	0.165	A	2.5	3	11
EV-0123	P04	DB	Lump Sum	7,898,039	764	4/15/2021	Variation	3069	1009.99	0.594	A	3.6	7	35
EV-0398	P11	DBB	Unit Price	14,908,494	820	5/15/2021	Errors in Emp. Req.	2218	1994	0	C	3.57	2	6
EV-0785	P21	DBB	Unit Price	13,628,604	659	2/5/2015	Variation	7128	965.66	0	D	2	3	23
EV-1243	P30	EPC	Lump Sum	59,453,073	1070	5/8/2020	Delays by Authorities	8134	2613.98	0.835	A	2.45	4	26

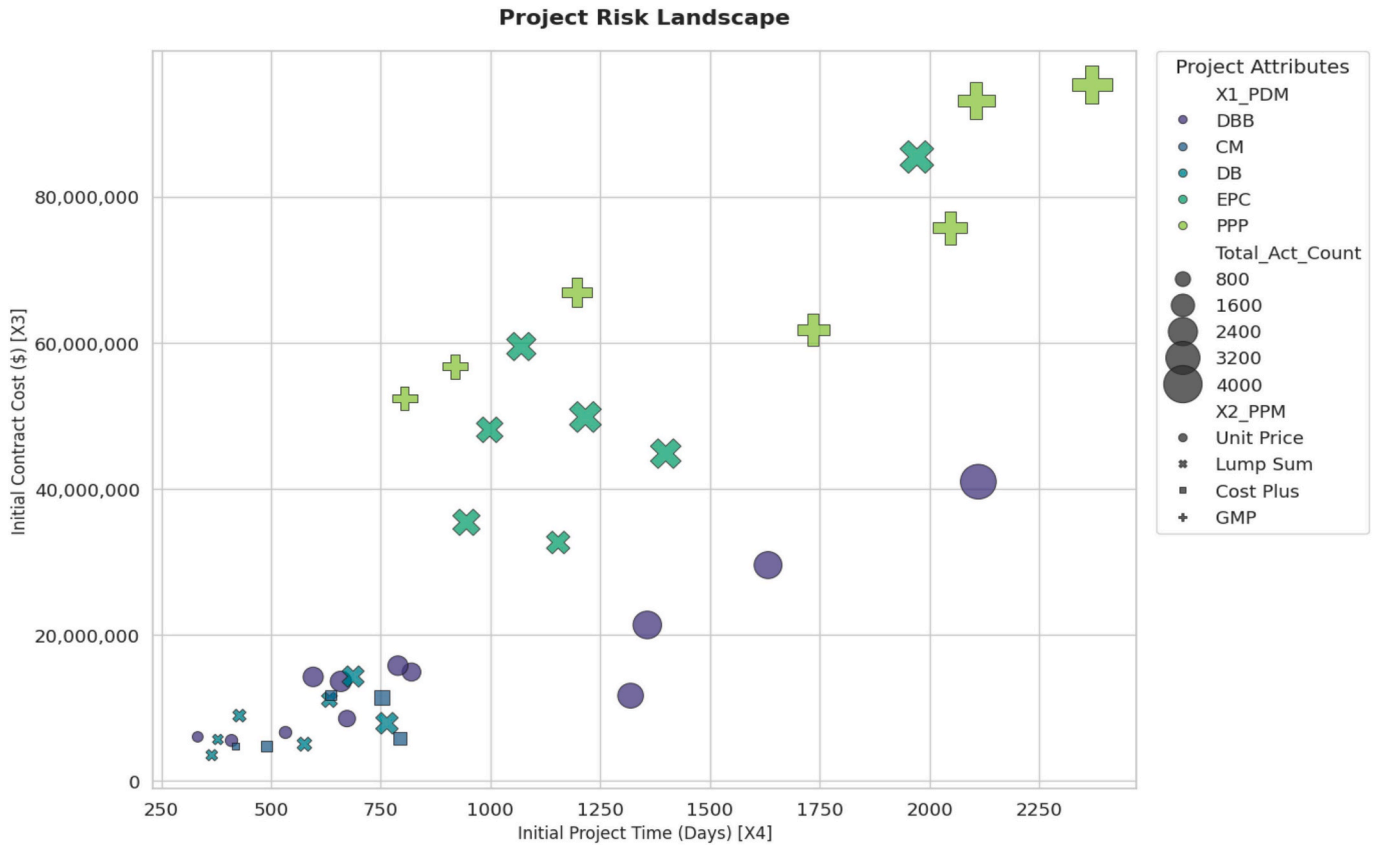


Fig. 4. Project risk landscape correlating cost, time, complexity, and delivery method.

resulting accuracy gain is often due to the collective wisdom of the ensemble reducing variance and bias. Ensemble methods are highly suitable for predicting the DDF due to several factors:

- Nonlinearity: The relationship between features like X_7 (ADM) and Y (DDF) is highly nonlinear, which ensemble trees naturally model effectively.
- High Variance Reduction: By averaging predictions across multiple learners, Ensemble methods provide greater robustness against noise and potential outliers present in the historical TIA analyzed dataset.
- Feature Interaction: The models inherently capture complex interactions, such as how the effect of a X_7 (ADM) and Y (DDF) is dependent on the level of X_{12} (PDD)

3.3.1. *Bagging algorithm*

The RF regressor is employed to provide a stable, variance reducing prediction and establish a qualified benchmark. RF operates through Bagging, constructing a multitude of decision trees during training where each tree is built from a random subset of the data sampled with replacement. A key feature of RF is the introduction of further randomness by selecting only a random subset of features (Random Subspaces) at each split, which substantially decorrelates the individual trees. The final DDF prediction is then determined by averaging the outputs of all trees in the forest. This process makes RF highly resistant to overfitting and produces intrinsically stable initial forecasts.

3.3.2. *Boosting algorithms*

Boosting algorithms are utilized sequentially to minimize bias and capture complex feature interactions, representing the state of the art in structured data prediction.

- XGBoost (eXtreme Gradient Boosting): Selected as the industry standard boosting framework, XGBoost is known for its speed, efficiency, and advanced regularization techniques. It constructs models sequentially, with each new tree correcting the errors of the preceding ensemble. Its formalized system for handling missing data and its ability to utilize parallel processing make it highly suitable for the scale and complexity of the TIA analyzed dataset.
- LightGBM (Light Gradient Boosting Machine): An optimized framework that improves upon traditional GBM by employing a novel approach to tree splitting. LightGBM grows trees' leaf-wise (as opposed to depth-wise), focusing on leaves that promise the greatest loss reduction. Furthermore, it utilizes Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to drastically reduce training time and memory usage, offering high speed while maintaining accuracy.
- CatBoost (Categorical Boosting): This algorithm is specifically designed for stability and robust generalization, utilizing a structure based on oblivious trees. While our categorical features are preencoded, CatBoost's methodology fundamentally reduces prediction shift and instability, making it highly effective for heterogeneous data that contains many categorical inputs.

3.3.3. *Stacking Ensemble*

The Stacking Ensemble is utilized as the final predictive mechanism to achieve maximum predictive performance. This advanced technique combines the outputs of the four base models (RF, XGBoost, LightGBM, CatBoost), which serve as the first level learners. The predictions generated by these base models are then fed as inputs to a single, second level meta learner which is Ridge Regression. The meta learner is trained to find the optimal weighted combination of the base models' outputs,

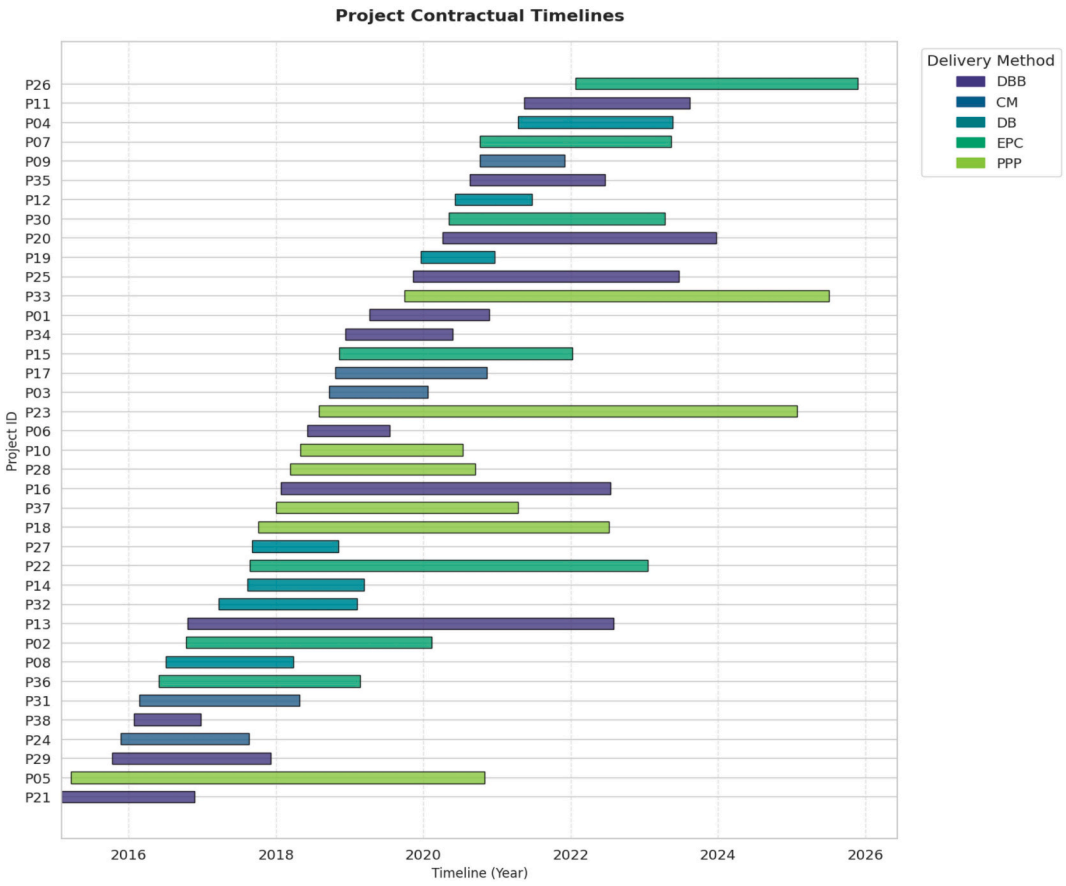


Fig. 5. Contractual timelines of the 38 analyzed projects grouped by delivery method.

3.4. Model evaluation

The predictive power of the model is validated by ensuring its generalizability to unseen data. The final curated dataset is subjected to a three-way partition:

- Training Set (70%): Used for fitting the model parameters during the primary learning phase.
- Validation Set (15%): Used for iterative hyperparameter tuning across all ensemble models.
- Testing Set (15%): Reserved exclusively for final performance evaluation to ensures the reported metrics reflect the model's true, unbiased accuracy on completely novel data.

3.4.1. Multicollinearity

The complexity of the feature set, particularly the engineered features, necessitates rigorous management of multicollinearity. For that, a comprehensive correlation matrix is computed for all continuous numerical features to identify pairs exhibiting strong linear relationships ($|r| > 0.90$). The primary goal is to eliminate the less influential feature from a highly correlated pair. This process ensures that the model coefficients remain stable, improving the predictive power of the individual remaining features and preventing the ensemble models from being misled by redundant information.

3.4.2. Overfitting prevention and sensitivity analysis

Since advanced Ensemble methods are utilized, a structured approach is essential to prevent the model from learning noise specific to the training set. The primary strategy involves k-fold cross-validation during hyperparameter tuning, where data is iteratively partitioned to optimize generalizability. To further rigorously address potential optimism bias arising from clustered data (multiple events per project), a secondary Group K-Fold sensitivity analysis is enforced. This protocol strictly isolates all events from specific projects during validation, ensuring the model predicts based on transferable causal relationships rather than memorizing project-specific idiosyncrasies.

3.4.3. Loss function and metric

The predictive performance is assessed using a suite of standard regression loss function. Primary loss function is Root Mean Square Error (RMSE). This metric is prioritized as it is highly sensitive to large errors, effectively penalizing prediction mistakes that would be critical in a construction project context (Eq. (5)).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (f(x_i) - y_i)^2}{n}} \tag{5}$$

The secondary loss function is Mean Absolute Error (MAE). This provides the average error magnitude without squaring the differences. MAE is highly interpretable for construction managers as it gives a direct estimate of the typical prediction error (Eq. (6)).

$$MAE = \frac{\sum_{i=1}^{n} |f(x_i) - y_i|}{n} \tag{6}$$

Furthermore, R^2 is chosen for metric. This measures the proportion of the total variance in the DDF that is explained by the engineered feature set. A high R^2 serves as final proof of the predictive quality of the methodological framework and the utility of the novel features (Eq. (7)).

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (f(x_i) - y_i)^2}{\sum_{i=1}^{n} (y_i - Y)^2} \tag{7}$$

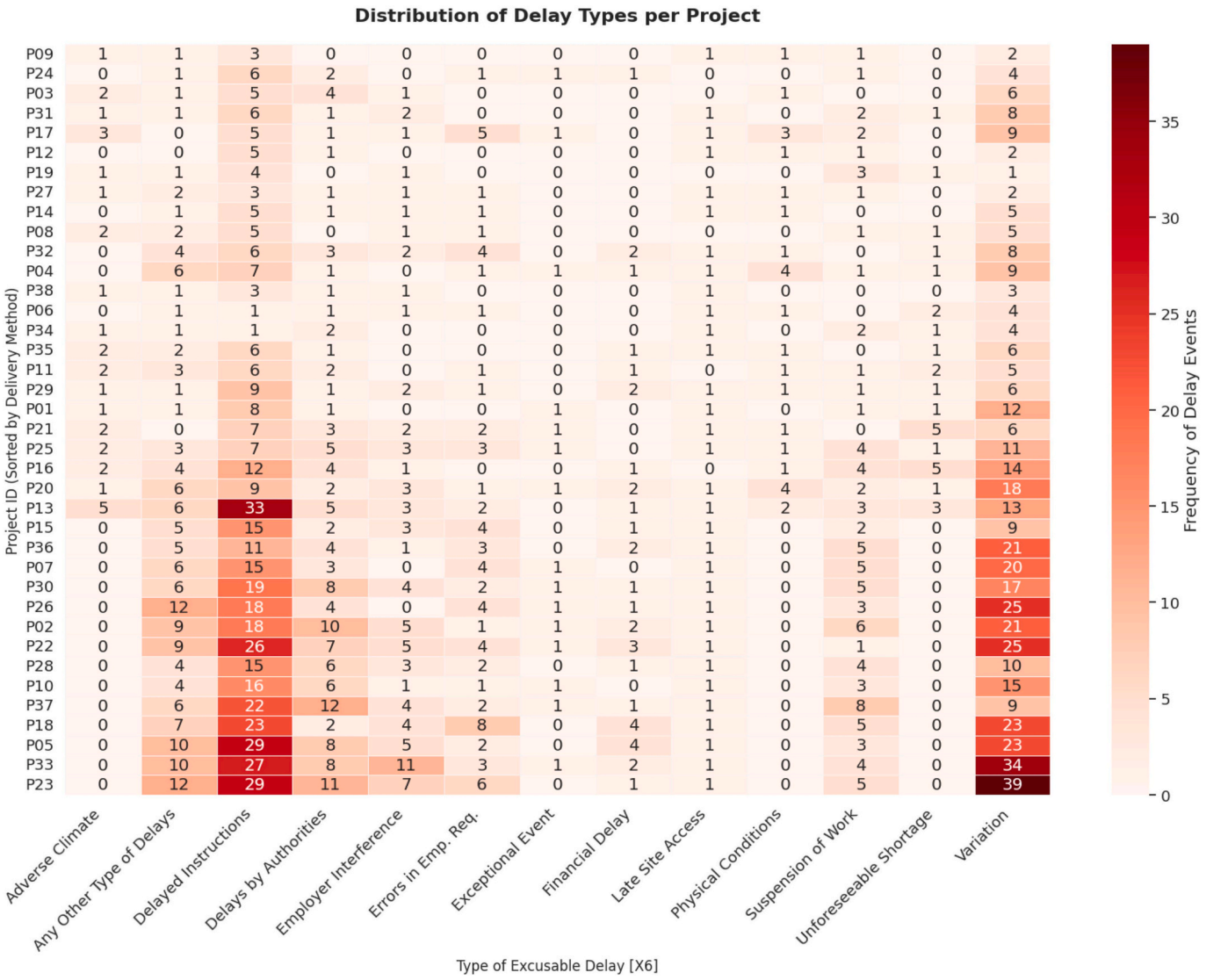


Fig. 6. Heatmap of delay type frequency per project.

3.5. MCDA interpretation

After training the model, it is vital to transform the opaque numerical output of the predictive models into a traceable context using an advanced interpretation layer. This challenge is addressed by integrating MCDA. Objective criteria weights are established using the SHAP framework applied to the RF surrogate model. Unlike traditional impurity-based importance, which can be biased toward high-cardinality features, SHAP values provide a game-theoretic measure of the marginal contribution of each feature to the model's output magnitude. The SHAP values (ϕ_{ij}) are first computed for each feature j and observation i. Global feature importance is then quantified using the mean absolute SHAP value across the training dataset according to Eq. (8).

$$I_j = \frac{1}{N}\sum_{i=1}^{N}\left|\phi_{ij}\right| \tag{8}$$

where I_j denotes the global importance of feature j, and N represents the total number of training observations. Since categorical variables are transformed using One-Hot Encoding during preprocessing, SHAP values are initially generated at the transformed feature level and subsequently aggregated back to their original feature groups to preserve interpretability at the contractual variable scale. The resulting global importance values are then normalized to generate the TOPSIS weighting vector via Eq. (9).

$$w_j = \frac{I_j}{\sum_{k=1}^{m} I_k} \tag{9}$$

where w_j represents the normalized TOPSIS weight of criterion j, and m is the total number of criteria. This normalization ensures that the sum of w_j must equal one. Furthermore, this study employs TOPSIS to translate predictive signals into a contractually defensible ranking of delay drivers. To strictly prevent interpretability leakage, a temporal partition protocol is enforced: global criteria weights (w_j) are derived exclusively from the training set via SHAP analysis and applied forward to evaluate unseen claim events in the test set. The fixed weighting vector w is subsequently injected into the TOPSIS framework for all downstream inference stages. The procedure follows four streamlined steps:

- Normalization: Inputs decision values (x_{ij}) are normalized to ensure scale invariance across heterogeneous metrics.
- Weighting: Normalized vectors are multiplied by the prefixed SHAP global weights via Eq. (10).

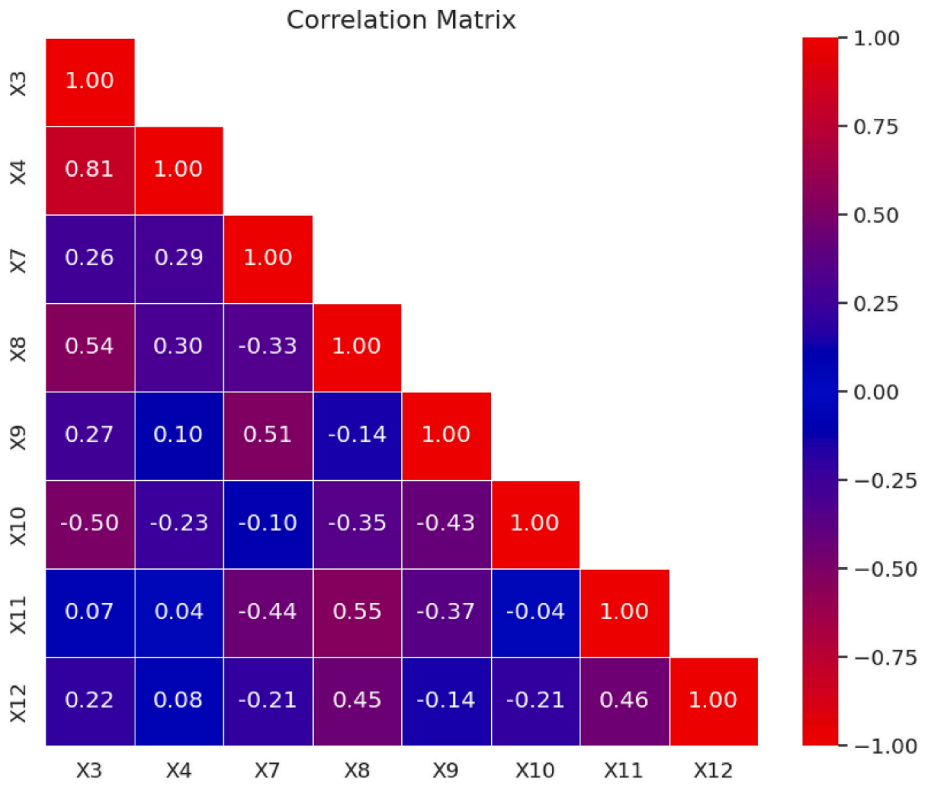


Fig. 7. Pearson correlation heatmap of numerical features.

Table 9
Optimized hyperparameters for individual base learners and the Stacking meta learner.

Model	Hyperparameter	Search Range	Optimized Value
RF	n_estimators	[100,300,800]	300
	max_depth	[5, 10, none]	none
	min_samples_leaf	[1,2,4]	1
	max_features	['sqrt', 'log2']	'sqrt'
XGBoost	n_estimators	[100,300,800]	800
	learning_rate	[0.01, 0.05, 0.1]	0.01
	max_depth	[1,3,5]	3
	subsample	[0.7, 0.9]	0.7
	colsample_bytree	[0.7, 0.9]	0.7
LightGBM	n_estimators	[100,300,800]	800
	learning_rate	[0.01, 0.05, 0.1]	0.01
	num_leaves	[15,30,50]	30
	min_child_samples	[5,10,20]	20
	reg_alpha	[0.01, 0.1, 1]	0.1
	reg_lambda	[0.01, 0.1, 1]	0.1
CatBoost	iterations	[100,300,800]	800
	learning_rate	[0.01, 0.05, 0.1]	0.01
	depth	[2,4,6]	4
	l2_leaf_reg	[1,3,5]	3
Stacking	Meta Learner	[Linear, RidgeCV]	RidgeCV

$$v_{ij} = w_j . r_{ij} \tag{10}$$

where r_{ij} denotes the normalized value of feature j for delay event i. This ensures the ranking logic is driven by the model's learned causal structure rather than post-hoc observation of the specific claim.

- Distance Calculation: Euclidean distances are computed.

Ranking: A Closeness Coefficient (CC_i) is generated ($0 < CC_i < 1$). The feature maximizing CC_i is identified as the dominant technical driver of the predicted DDF.

4. Results and discussion

4.1. Data Description and statistical analysis

The dataset encompasses a diverse range of project scales and contractual risk profiles, with ICC (X_3) ranging from $3.5 million to $95.3 million, IPT ($X_4$) spanning 333 to 2,370 days, and CSD (X_5) ranging from 2015 to 2025. The portfolio represents a heterogeneous mix of infrastructure sectors, comprising heavy civil (42%), high-rise commercial (26%), industrial/power (18%), and transport infrastructure (14%). To validate the representativeness of the training data, Fig. 3

Table 10
Comparative performance metrics of Ensemble models.

Model Architecture	Algorithm	RMSE				MAE				R2	
		Normalized		Original Unit (Days)		Normalized		Original Unit (Days)			
		Train	Test	Train	Test	Train	Test	Train	Test	Train	Test
Stacking Ensemble	Meta Learner (Ridge)	0.344	0.445	8.1	10.5	0.273	0.315	6.4	7.4	0.889	0.817
Boosting	XGBoost	0.356	0.449	8.4	10.6	0.287	0.32	6.8	7.5	0.88	0.813
	LightGBM	0.359	0.451	8.4	10.6	0.289	0.321	6.8	7.5	0.877	0.811
	CatBoost	0.361	0.456	8.5	10.8	0.298	0.324	7	7.6	0.873	0.807
Bagging	RF	0.381	0.491	9	11.6	0.33	0.335	7.8	7.9	0.868	0.774

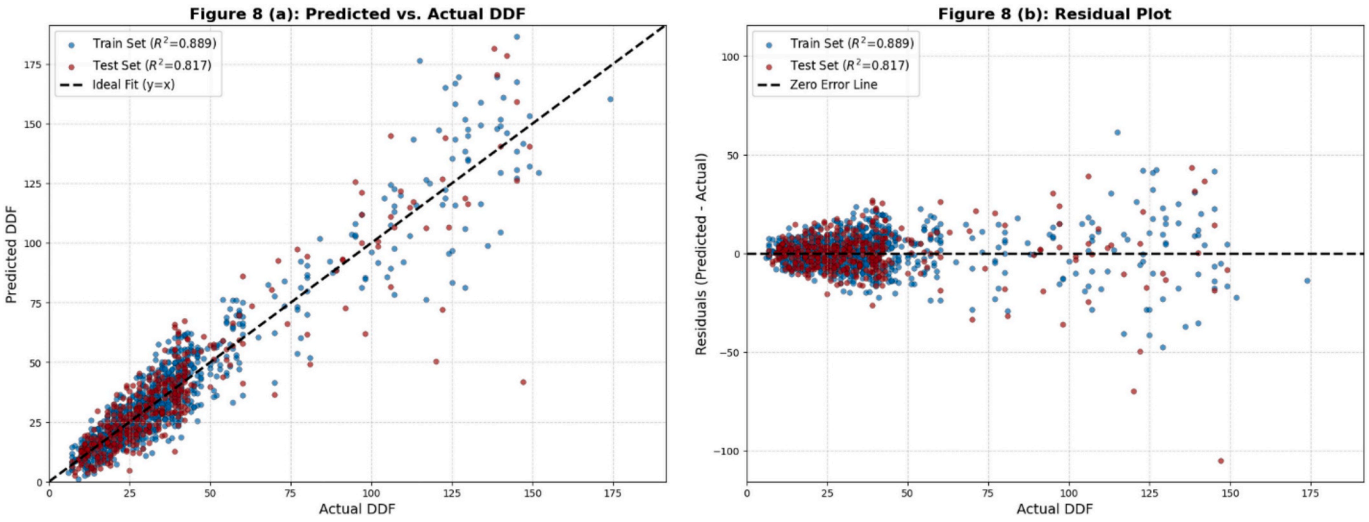


Fig. 8. Final model evaluation diagnostics. (a) Scatter plot of predicted vs. actual DDF. (b) Residual error plot.

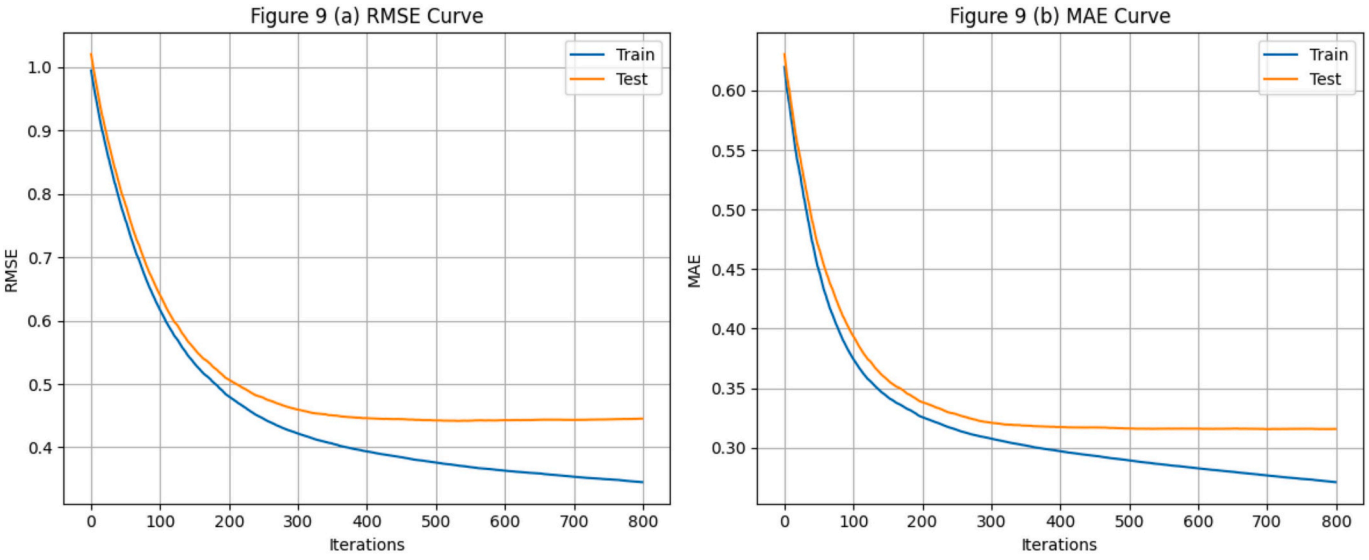


Fig. 9. Stacking Ensemble learning curves. (a) RMSE (b) MAE.

illustrates the distribution of numerical features and Table 7 presents the comprehensive distribution of the primary categorical features.

Table 8 provides a representative sample of the final training dataset, displaying the engineered features (X_7 – X_{12}) alongside the other variables for specific delay events.

To visualize the interaction between project scale, complexity, and contractual risk, Fig. 3 presents the project risk landscape. This five-dimensional scatter plot correlates IPD and ICC with the PDM, PPM and total activity count. As shown in Fig. 4, there is a distinct clustering of high-risk profiles. PPP and EPC projects consistently occupy the upper right quadrant, indicating high cost and long duration, and are associated with larger bubble sizes, denoting higher schedule complexity. Conversely, DBB and CM projects cluster in the lower left, representing lower capital intensity but a wide variance in duration.

The temporal validity of the dataset is confirmed in Fig. 5, which illustrates the contractual timelines of all 38 projects. The Gantt chart visualization demonstrates that the dataset covers a contiguous execution period from 2015 to 2025, ensuring the model is trained on data reflecting diverse economic conditions.

The frequency and distribution of specific delay causes exhibit a clear risk signature unique to each contract type. Fig. 6 presents a heatmap of delay event frequency per project. The heatmap reveals critical validation of the FIDIC silver book risk transfer rules: projects under EPC and PPP delivery methods show zero frequency for “Unforeseeable Physical Conditions” and “Adverse Climatic Conditions,” confirming that these risks were successfully transferred to the contractor and thus did not generate excusable delay events in the training set. In contrast, “Delayed Instructions” and “Variations” appear as high intensity bands across all project types, identifying them as universal drivers of schedule disruption.

Fig. 7 presents the Pearson Correlation Matrix for the primary numerical features. The analysis confirms that the feature engineering process successfully created distinct, nonredundant signals for the model.

- Independence of Magnitude and Criticality: A crucial validation is the low correlation between ADM (X_7) and CTR (X_9). This confirms that the size of a delay event is independent of its criticality; a small delay can be highly critical, just as a large delay can be noncritical. This distinction is vital for the model to learn that large does not automatically mean urgent.

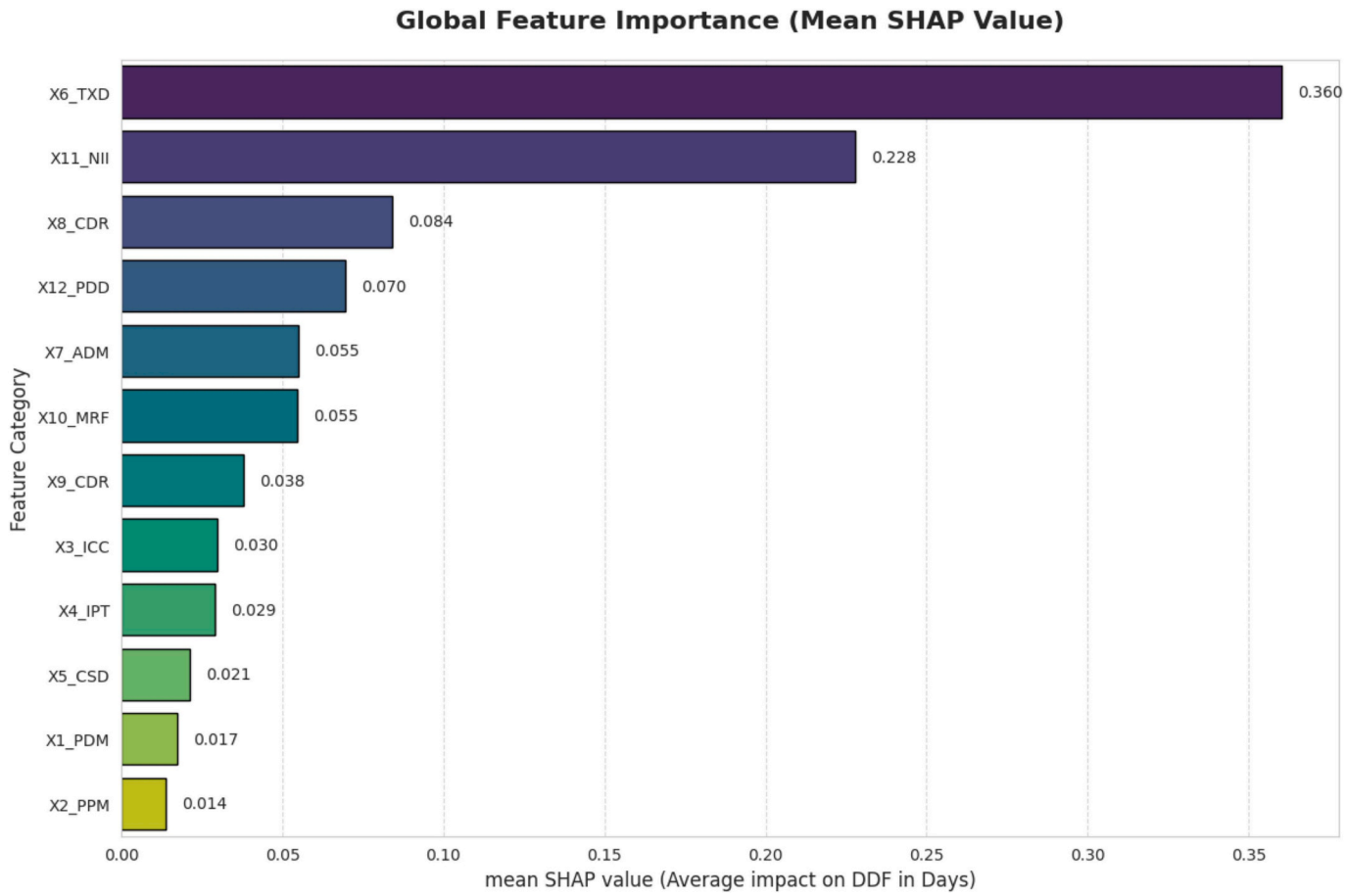


Fig. 10. Global feature importance plot.

- Constraint vs. Complexity: The relation between MRF (X_{10}) and NII (X_{11}) is weak, indicating that schedule complexity and schedule tightness are separate risk dimensions. High interconnectivity does not necessarily imply zero float, allowing the model to weigh them separately.

The maximum observed correlation coefficient between any pair of engineered predictors is below the threshold of $|r| = 0.81$. Consequently, no features required elimination due to multicollinearity.

4.2. Model performance and validation

The superior predictive performance of the Ensemble architecture was achieved through a rigorous hyperparameter tuning process. A grid search strategy coupled with 5-fold cross validation was executed for each base learner to identify the configuration that minimized the RMSE while maximizing generalizability. The tuning process focused on finding the optimal balance between model complexity and regularization. Table 9 presents the final set of optimized hyperparameters for all five models used in the study.

The selection of a relatively low learning rate paired with a high number of estimators for the boosting algorithm indicates that the models required a gradual learning process to effectively map the complex interactions between features and target. Conversely, the RF stabilized earlier, reaching optimal performance with 300 trees, confirming its role as a lower variance, highly stable benchmark.

The predictive performance of the candidate models was evaluated on the unseen test set. Table 10 presents the comparative results. The analysis confirms that the Stacking Ensemble achieved the highest predictive accuracy across all metrics, validating the hypothesis that combining diverse learning strategies (Bagging and Boosting) yields superior generalization for complex construction data.

Models were ranked based on the primary metric, RMSE, which penalizes large prediction deviations. The Stacking Ensemble demonstrated a 9.5% reduction in RMSE compared to the single best boosting model (XGBoost) and a 23.9% reduction compared to the RF. This performance superiority is attributed to the meta learner's ability to dynamically weight the base models:

- Handling Categorical Complexity: CatBoost provided superior stability for rows with complex categorical combinations such as X_6 interacting with X_1.
- Handling Nonlinearity: XGBoost and LightGBM excelled at modeling the nonlinear thresholds in the engineered features, such as the tipping point where X_9 shifts a delay from noncritical to critical.

To address the potential for optimism bias arising from clustered data (where multiple delay events originate from the same project), a secondary group K-Fold Cross-Validation was performed. Unlike the standard random split, which treats all events as independent, this validation enforces a strict "Leave-Projects-Out" protocol where all delay events from a specific project are held out for testing, ensuring the model is evaluated on entirely unseen project environments. The analysis yielded a Project-Grouped R^2 of 0.787, compared to the random-split test R^2 of 0.817. The marginal performance drops of only 2.45% confirms that while project-specific idiosyncrasies contribute minor variance, the model relies primarily on the generalizable engineered features rather than memorizing specific identities. This validates the robustness of the DDF construct as a transferrable metric across diverse infrastructure assets, capable of predicting delay magnitude on new

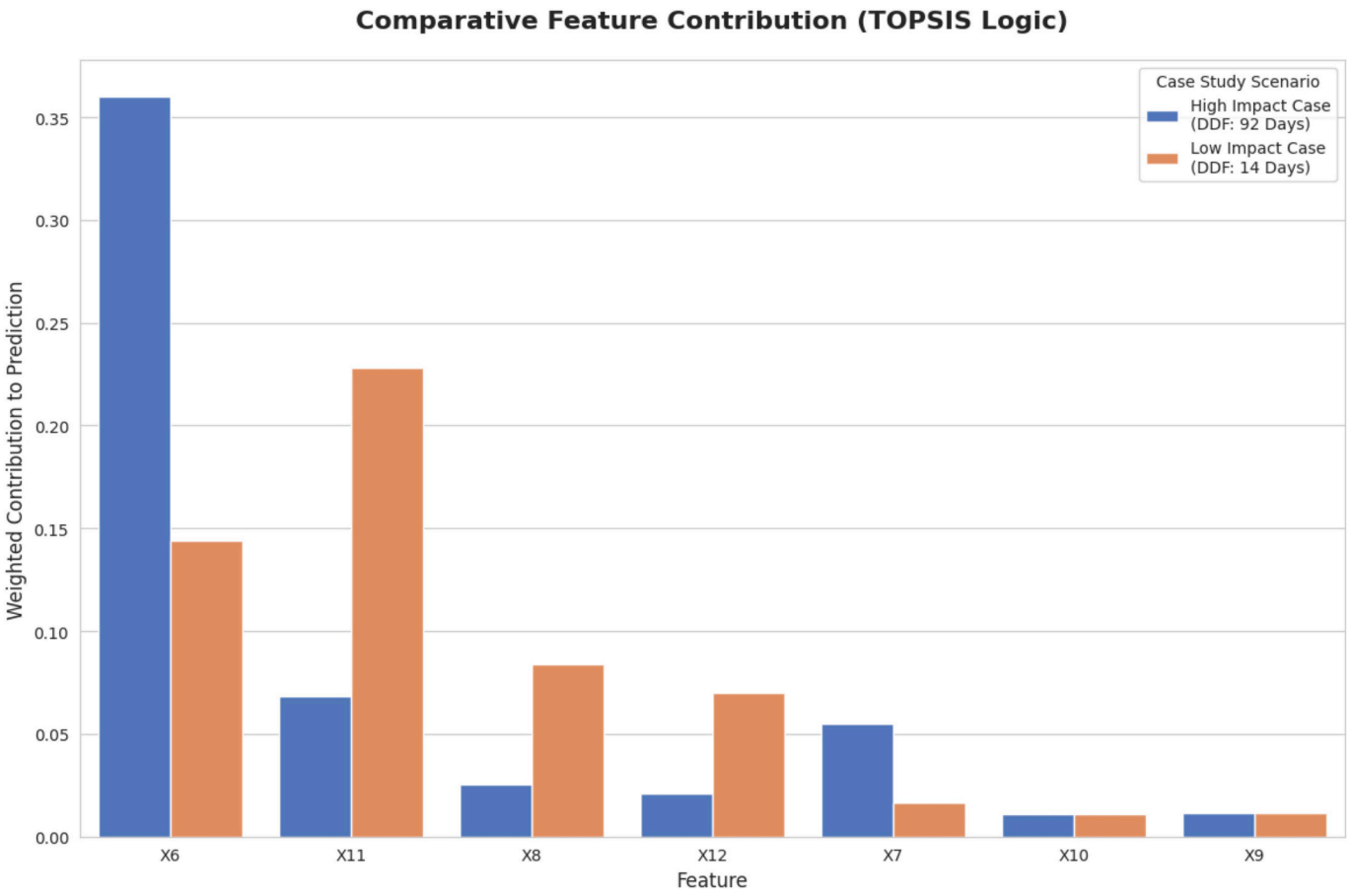


Fig. 11. Comparative feature contribution for case study A vs. case study B.

projects without prior exposure to their specific baseline logic.

Fig. 8 presents the diagnostic plots used to assess the best model's reliability and error characteristics. Fig. 8(a) illustrates the correlation between the predicted DDF and the actual DDF. The data points exhibit a tight clustering along the 45-degree diagonal line ($y = x$), indicating a high degree of predictive accuracy across the full range of delay magnitudes. The model shows exceptional precision in the lower-left quadrant, accurately forecasting routine operational delays like delayed instructions with minimal variance. Crucially, the model maintains its linear trajectory even for extreme delay events (upper-right quadrant), avoiding the common saturation bias where models tend to under predict catastrophic events. This confirms that the Stacking Ensemble successfully integrates the nonlinear signals from the boosting learners to capture the full scale of project risk.

Fig. 8(b) displays the Residual Plot (Prediction Error vs. Predicted Value). The residuals are randomly distributed around the horizontal zero axis, demonstrating homoscedasticity (constant variance). There is no discernible pattern like fanning or curvature in the residuals, confirming that the model does not systematically over or under estimate delays based on their size. While a few outliers exist (represented by points deviating further from the zero line), they are symmetric and sparse, indicating that the model is robust to noise and does not suffer from significant structural errors.

To confirm the stability of the meta learning process and assess the potential for overfitting, the learning curves for the Stacking Ensemble were generated. Fig. 9 illustrates the progression of the primary (RMSE) and secondary (MAE) metrics. As shown in figure, both the train and test error converge. The narrow gap between the training curve and the testing curve indicates that the Stacking Ensemble generalizes well to unseen data and does not suffer from high variance (overfitting). The plateauing of the curves suggests that the model has successfully captured the maximum extractable signal from the available features.

4.3. MCDA interpretation

The first step in the interpretation framework was to establish the global hierarchy of delay drivers using SHAP values extracted from the RF surrogate model. Fig. 10 presents the global importance ranking of the twelve feature categories based on their marginal contribution to the model's output.

Consistent with construction claims theory, the TXD (X_6) remains the most predictive factor. The model confirms that the inherent technical severity of the root cause establishes a baseline duration that dictates the DDF range regardless of project variables. A critical finding of this study is the high importance of the complexity compared to magnitude. The model indicates that the connectivity of the affected work is a far stronger driver of delay time than the sheer volume of the work itself. This suggests that administrative and coordination friction in complex network logic creates longer delays than simple, large volume linear work. CDR (X_8) emerged as the third most important driver. This confirms that high intensity activities (high cost per day) require significantly more time to reprocure or redesign, directly inflating the DDF. PDD (X_{12}) plays a notable role, validating the hypothesis that project distress is cumulative; a delay occurring in an unstable environment tends to be resolved more slowly than one in a stable period.

Surprisingly, schedule constraints such as CTR (X_9) and MRF (X_{10}) lower importance relative to complexity. This implies that while zero float conditions dictate when the project finishes (EOT), DDF is driven more by the complexity of the fix than the tightness of the schedule. Consistent with the study's methodology, static project data (initial cost

and time, contract type) exhibited negligible importance. This underscores that DDF is a dynamic, event specific phenomenon that cannot be accurately predicted using macro level project characteristics alone; granular, specific data is mandatory.

The TOPSIS transforms the global feature weights into a local ranking of drivers for any single prediction. Two contrasting case studies from the test set were analyzed to demonstrate the model's capacity for nuanced reasoning:

Case Study A: The "High Impact" Scenario.

- Event Profile: A "Suspension" event on an EPC project with a predicted DDF of 92 days (High Severity).
- TOPSIS Analysis: The analysis reveals that the TXD (X_6) is the overwhelming contributor, with a weighted score of 0.37. In contrast, other potential drivers such as NII (X_{11}) and PDD (X_{12}) made negligible contributions to the ranking distance.
- Contractual Interpretation: The model correctly identified that for this specific event, the severity was dictated by the contractual definition of the delay type rather than the complexity of the schedule logic or project stability. The intrinsic nature of the event established a high baseline duration that other factors could not mitigate.

Case Study B: The "Low Impact" Scenario.

- Event Profile: A " Delayed Instructions" event on a DBB project with a predicted DDF of 14 days (Low Severity).
- TOPSIS Analysis: In this scenario, NII (X_{11}) contribution was notably high (0.23), significantly exceeding its contribution in the high severity case. However, the contribution of the TXD (X_6) dropped drastically to 0.14.
- Contractual Interpretation: This contrast provides a sophisticated insight, even though the delay occurred in a complex area of the schedule logic, the routine nature of the delay cause prevented the duration from escalating. The model successfully reasoned that complexity alone does not create a catastrophic delay if the root cause is minor, effectively suppressing the predicted DDF to just 14 days.

Fig. 11 visualizes the TOPSIS contribution scores for these two cases side by side, illustrating how the model weighs features differently depending on the specific context of the event.

4.4. Practical Implications for contract Management

The dominance of TXD as the primary predictor confirms that the contract's specific definition of "Excusable Delay" is indeed the governing factor in determining time entitlement. However, the significant influence of X_{11} and X_8 highlights a critical nuance often overlooked in standard contract clauses: the technical complexity of the change determines the duration of the extension, independent of the legal liability. This suggests that claims assessment under Sub-Clause 20.1 should explicitly account for the density of the affected work, not just its critical path status.

Contractors can utilize the proposed predictive framework to substantiate their initial notices of claim. By inputting the event parameters into the model, a contractor can generate a scientifically backed baseline for the DDF. Submitting a claim supported by a probabilistic forecast reduces the perception of optimism bias. Furthermore, the TOPSIS analysis demonstrates that for complex delay events, contractors should explicitly document the NII (X_{11}) in their narrative to justify longer extension requests, as the model proves this is a valid driver of time consumption. For consultants and contract administrators the model serves as an objective benchmarking tool. Upon receiving a claim, the consultants can run the event parameters against the model. A significant deviation between the contractor's claimed DDF and the model's predicted DDF acts as an early warning signal for potential inflation or underestimation. The strong predictive power of X_8 empowers consultants to challenge claims that lack sufficient resource loading data, shifting the discussion from abstract time estimates to concrete resource per day metrics.

5. Conclusion

This study addressed the persistent challenge of subjectivity and inefficiency in the assessment of delay claims within the construction industry. By integrating forensic schedule analysis with advanced ML, the research successfully developed a predictive framework for forecasting the DDF in projects governed by the FIDIC 2017 Suite of Contracts. Key findings of this study include: (1) The technical duration of a delay event is not a random stochastic variable, but a deterministic outcome driven by identifiable schedule mechanics and event characteristics. (2) The Stacking Ensemble (combining RF, XGBoost, LightGBM, and CatBoost) proved to be the optimal architecture, achieving a Test R^2 of 0.817 and a RMSE of 0.445, demonstrating enhanced predictive performance relative to the individual baseline models. (3) SHAP analysis overturned the assumption that project size dictates delay magnitude; instead, the type of excusable delay (X_6) was identified as the dominant baseline driver, while network interdependency (X_{11}) and cost density ratio (X_8) served as the critical multipliers. (4) The integration of the TOPSIS framework successfully transformed the "black box" ensemble model into a transparent decision support tool, providing contractually defensible reasoning by isolating specific drivers for individual predictions. (5) The framework aligns with the FIDIC 2017 by empowering consultants with an objective validation mechanism to rapidly identify inflated claims and enabling contractors to submit scientifically grounded initial notices.

While the dataset contains 1,594 delay events, these observations were derived from a limited number of projects within a relatively concentrated regional context, which can constrain the broader generalizability of the learned predictive relationships. In addition, although the proposed framework demonstrates robust predictive accuracy, it is bounded by its specific calibration to FIDIC contracts, potentially requiring recalibration for generalizability to other standard forms. Furthermore, the current framework focuses exclusively on quantitative technical drivers, excluding negotiation variables. Future research should address these boundaries by integrating NLP to ingest unstructured correspondence.

CRediT authorship contribution statement

Mohammad Taghipour: Writing – original draft, Software, Investigation, Formal analysis, Data curation. **Mostafa Khanzadi:** . **Abdollah Amirkhani:** Writing – review & editing, Supervision, Software, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Mohammad Taghipour is a Ph.D. candidate in Construction Management at Iran University of Science and Technology (IUST), Tehran, Iran, where he also received the M.Sc. and B.Sc. degrees in construction management and civil engineering, respectively. He has over five years of professional experience in construction project planning and control, delay and claim analysis, and technical office management. His research interests include construction project management, delay prediction and claims, BIM-based project control, and machine learning applications in construction.



Mostafa Khanzadi received the B.Sc. and M.Sc. degrees in civil engineering from the University of Manchester, U.K., and the Ph.D. degree in civil engineering from Hokkaido University, Japan. He is currently a Professor in the Department of Civil Engineering at Iran University of Science and Technology (IUST), Tehran, Iran. His research interests primarily include construction and project management, construction risk analysis, system dynamics, strategic and financial management of construction projects, contract management, and decision-support systems. He is a member of the American Concrete Institute and the Japan Concrete Institute.



Abdollah Amirkhani (Senior member, IEEE) received the M. Sc. and Ph.D. degrees (Hons.) in electrical engineering from Iran University of Science and Technology (IUST), Tehran, in 2012 and 2017, respectively. He is currently an Associate Professor with the School of Automotive Engineering, IUST. He has been actively involved in several national research and development projects, related to the development of new methodologies and learning algorithms based on AI techniques. His research interests include machine vision, fuzzy cognitive maps, data mining, and machine learning. He received the Outstanding Student Award from the First Vice President of Iran, in 2015. In 2016, he was conferred award by the Ministry of Science, Research and Technology. He is an Associate Editor of the Engineering Science and Technology, an International Journal.